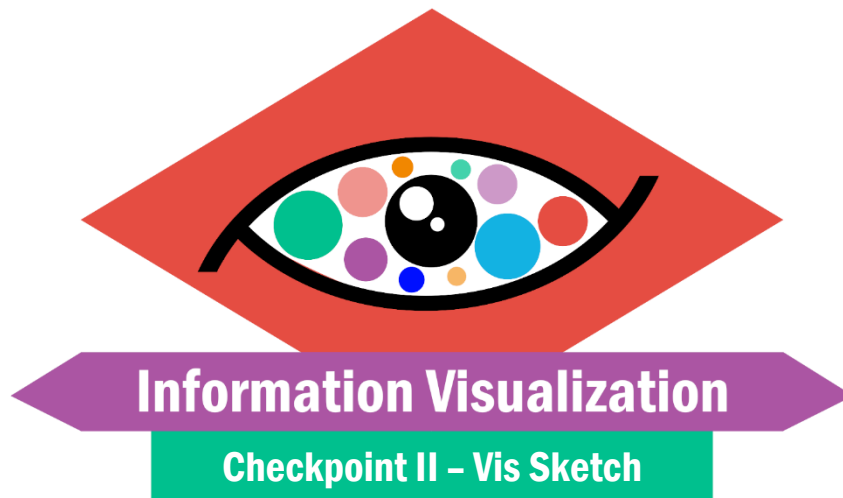




Objective

At the end of this checkpoint, you must have a sketch depicting the main aspects of your visualization, which you can then implement.

Requirements

Create a paper-and-pencil sketch of your visualization, based on the Task and Data Abstractions you defined in Checkpoint I. Your prototype must incorporate at least three distinct but interconnected visual idioms. Note that using more than three visual idioms will not result in a higher grade.

Keep in mind that some tasks should be supported by combining multiple idioms to leverage their synergies. Avoid selecting idioms that are only capable of answering the specific questions you defined in CP I. A well-designed dashboard should support data exploration, enabling users to answer not just your initial questions, but many others as well. Think of the questions as a starting point, not a design constraint.

These sketches may be done using prototyping tools, but we want you to create a low-fidelity prototype. You must focus on choosing the visual encodings for your data types and the idioms for supporting your questions instead. Hand drawings are preferred ***if we can understand the depicted visualization techniques***. Consider **whether the idioms you choose are the most effective for what you want to show** (datatypes, tasks, scalability, etc.) and present adequate justifications for their choice based on those aspects. There are dozens of visual idioms available, each with distinct advantages and disadvantages.

Please create one sketch for each visual idiom, plus an additional sketch showing the overall dashboard layout where these idioms will be integrated. Use the individual sketches to emphasize specific details of each idiom and use the dashboard overview sketch to provide a clear sense of how all the components fit together. Pay close attention to your layout, proportions, alignments, and relative sizing. We will evaluate these aspects carefully, so simply placing visualizations side-by-side without thoughtful arrangement will result in a low grade.

If your dashboard includes interactivity that changes visual representations or states, consider including multiple sketches to demonstrate these variations. Showing this level of detail will demonstrate deeper understanding and will help you earn a higher grade.

For each visual idiom, clearly describe the marks and visual channels you are using, along with the rationale for your choices. Ground your explanation in the theoretical concepts covered in class, particularly considering the task types and data types you selected in Checkpoint I. Again, avoid relying

on common sense alone. Demonstrate a thoughtful, theory-based decision process.

Additionally, discuss alternative visual idioms you considered, explaining their respective advantages and disadvantages. This will show that you have carefully thought about which visual idioms are the best fit for your data and tasks.

Be sure to include all interaction techniques supported by your chart (e.g., hover, click, drag, brush), and describe how the chart should respond both visually and internally to these user interactions. We will assume that standard interactions typical for specific visual idioms will be included, so explicitly mention any additional or custom interactions. ***Do not forget to incorporate legends and scales!***

You must discuss the **dimensionality** as well. For instance, consider that you have a categorical and a ratio variable. You could use a bar chart to show a distribution. But what if you have four ratio variables? Should you use a stacked bar chart? Small multiples? Will you consider interaction if the user does not need to see them all at the same time? Will there be animations to show some evolution? Another relevant example is when you consider temporal dimensions. Linear/cyclic, hierarchic or not, spans or intervals or instants... All these features help you choose the more adequate visual idiom. A similar logic applies to visualize geolocations: what projection will you use? Which frontiers? Be precise. We will discuss with your group the design choices and you must be able to justify them.

Also, you must show us how the several idioms you chose, working *in tandem*, will address your needs. Present how the idioms are **integrated**, e.g., if you click on a mark in the scatterplot, the mark is highlighted in the line chart through a change in the color fill from black to blue. Bottom-line: show how the integration helps the user to complete the tasks and generate insights that go beyond a single visualization. To that end, you must provide a **step-by-step guide to answer each question from Checkpoint I** using your visualization dashboard, no images needed. However, you must create a **storyboard showing how to answer at least one of the questions from Checkpoint I**. You can choose the question you think is a better fit for the storyboard, but answering it **must require the combined use of more than one of the visualization idioms** (e.g., finding something in one view, selecting it, causing that item to be highlighted in another view, where you then compare it with others, etc.).

Again, remember that you do not need (should not) limit yourself to the tasks/questions in Checkpoint I if there are obvious opportunities to explore the data better, given your choice of encodings/idioms. Beware of ***missed opportunities***.

Example

Remember when we chose the scatterplot to visualize the correlation between the wines' prices and ratings? We opted for a scatterplot since it allows us to examine the overall pattern of the data in terms of direction, form, and strength. It also helps us to easily identify outliers. The marks were circles, with the x-axis channel encoding the price and the y-axis channel encoding the rating. However, we added an additional mark: a line to represent the strength of the association.

Other alternatives could be a KDE plot (to show the density distribution through a contours) or a hexbin plot (like a heatmap to bin the chart area into hexagonal regions). However, both charts suffer from the lack of individual marks to demark the relationship between the two variables. Instead, the KDE and the hexbin plots are better to see the frequency with which the individual values appear in areas/bins.

You could also use single-variable charts, such as bars or dot plots, to depict the price and the ratings. However, a striking drawback hinders the analysis: the fact that these charts require the marks to be ordered in an axis prevents the user from grasping the correlation visually.

More examples? Parallel coordinate plots can help visualize this relationship, but they are overkill for

only two variables. The appeal to use parallel coordinate plots is the ability to switch axis positions to view specific relationships more in-depth.

Deliverables

Create a **5-page document using the provided template** and submit it online, until two days before your class (ex: classes on Monday must submit until Friday end of day). Name your deliverable as '**CPII-
<group number>.pdf**'. For instance, 'CPII-02.pdf'.

Penalties

- Documents over 5 pages long: **1 grade point penalty per extra page.**
- Document uploaded after the deadline: **0.5 grade points penalty per hour of delay.**
- Incorrect file type or name: **0.5 grade points penalty per file or name.**
- Document template altered (wider margins, smaller font, etc.): **1 grade point penalty.**
- Small images or with low resolution: **0.5 grade points penalty.**

Constraints

Think about the visualization idioms you are going to use. Your project needs to contain *at least* three different linked visualization techniques. **You do not get extra points for effort if you create more than three idioms.** However, **you will be deducted if you try to force everything into just three and this does not make exploring the data effective, efficient and clear.** Sometimes it is better to implement a fourth idiom that helps answering or complementing a data scope/question. Keep that in mind and make sensible decisions that you can justify based on theoretical and practical knowledge.

At least one visual idiom must be in the whitelist that is available in Moodle. Go check it!

Tasks to perform during the lab

The professor will provide feedback. The grade will be made known one week later (see below). If you are not receiving feedback, then you must be peer assessing your colleagues' submissions.

Grading

Your work will be graded according to the following parameters:

- **Overall organization (5%):** carefully craft the dashboard layout. Beware of white space.
- **Visual Idioms and their descriptions (40%):** carefully present each visual idiom and their features. Discuss alternatives.
- **Interaction (15%):** describe the interaction of each chart (e.g., hovers, clicks, brush).
- **Integration (15%):** present the integration between all charts.
- **Articulation with Checkpoint I (10%):** the visual idioms support the questions from Checkpoint I.
- **Missed opportunities (10%):** identify opportunities that were not visible in Checkpoint I but now can be achieved, or extensions to your proposal that are obvious (e.g., adding a fill encoding to a scatterplot to depict a third variable).
- **Peer assessment (5%):** grade your peers' submissions in the class.

An important note on grading for this and all other Checkpoints: always *justify* your choices, based on the basic principles you have learned in this course (adequacy of channels to data types, human perception, etc.). Don't just describe what it is, but especially *why* it is as it is.

Additional Notes

After you deliver your document, your work will be graded. HOWEVER, this grade **can be improved by up to two grade points** if you correct any faults pointed out by the professor and submit a revised version of the document HIGHLIGHTING THOSE CHANGES up the beginning of the class taking place 7 days after you receive feedback in class. ***Only highlighted changes will be considered.***